6G & Future Networks

A3-PS016-TC243

Simulation-Based Handover Decision Evaluation for High-Mobility Users

Team LostPacket

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AIORI Hackathon 3 · October 14, 2026

Abstract: Problem Description

The problem

A moving phone must switch from one cell to the next (handover). At high speed the timing is hard: late switches drop the link, early or repeated ones cause ping-pong.

Today's baseline

The conventional A3 rule switches when a neighbour cell's signal is stronger than the serving cell's by a margin.

Our question

Can one supervised learned policy beat the A3 baseline for users moving at two different speeds across three cells?

Scope

Simulation only, in a reproducible ns-3 LTE setup. No 6G conformance claim, new radio stack, deep RL or field deployment.

3

LTE cells (eNBs)

2

user speeds

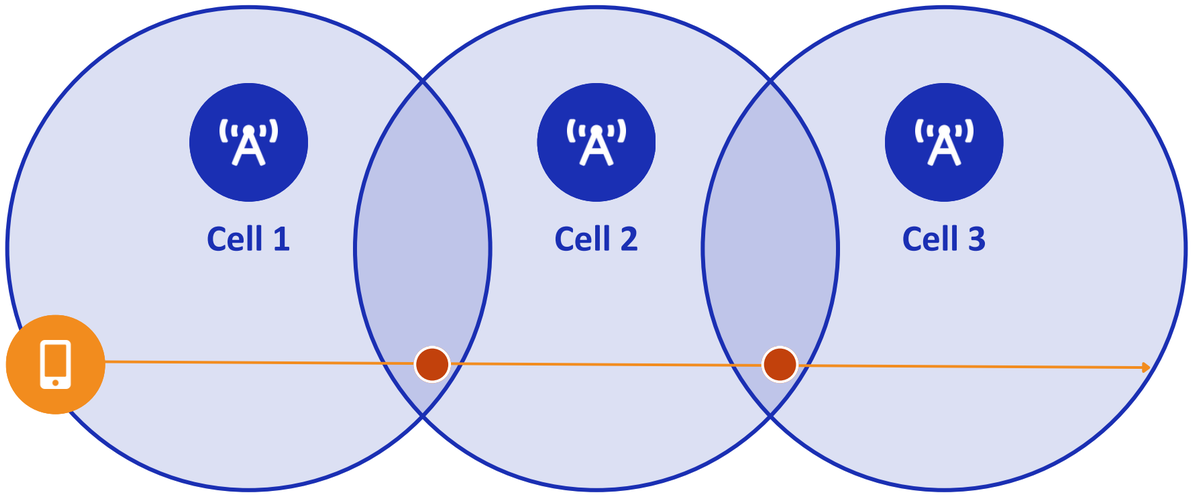
1

learned model vs A3 baseline

4

metrics, over multiple random seeds

Background



Handover zones
(Overlapping Regions)

30 km/h

More time to decide

120 km/h

Little time, so higher risk

Key terms

RSRP: (Reference Signal Received Power) measure how strong the radio signal is from a specific base station

A3 event: neighbour RSRP is better than serving RSRP by an offset (with hysteresis) for a set time-to-trigger [2][3].

Too late: when the link fails before the handover completes, a handover failure.

Too early or repeated: the phone bounces between cells (ping-pong) and each switch interrupts data.

High speed: signals changes quickly, leaving little time to decide.

The main goal is for a handover system to pick the right moment and the right target cell, so a handover neither fails nor repeats.

Literature & RFC Review

RFC 2330: Framework for IP Performance Metrics [1]

Repeatability. Scenario files and random seeds are fixed and logged, so every run can be reproduced.

Singleton, sample, statistic. One run is a singleton, all seeds form a sample, and we report the mean or median with the sample size N.

Rates, not probabilities. Metrics are counts and rates, such as failures per handover.

Document uncertainty. We report the spread across seeds and the limits of the simulator.

3GPP TS 36.331 and TS 36.300

Define the A3 event (offset, hysteresis, time-to-trigger) and the LTE handover procedure. This is the basis of our baseline [2][3].

ns-3 LTE module

Open-source LTE/EPC simulator with eNBs, UEs, mobility models and a built-in A3-RSRP handover algorithm. This is our single simulator [4].

Proposed Solution

1

Simulate

ns-3 LTE with 3 eNBs. A user moves across the cells at 30 and 120 km/h.

2

Baseline

A3 rule with fixed offset, hysteresis and time-to-trigger.

3

Collect

Log RSRP, speed and handover outcome in every run.

4

Learn

Train one random forest (scikit-learn) to decide when to hand over [5][6].

5

Evaluate

Compare both policies over many seeds on the four metrics.

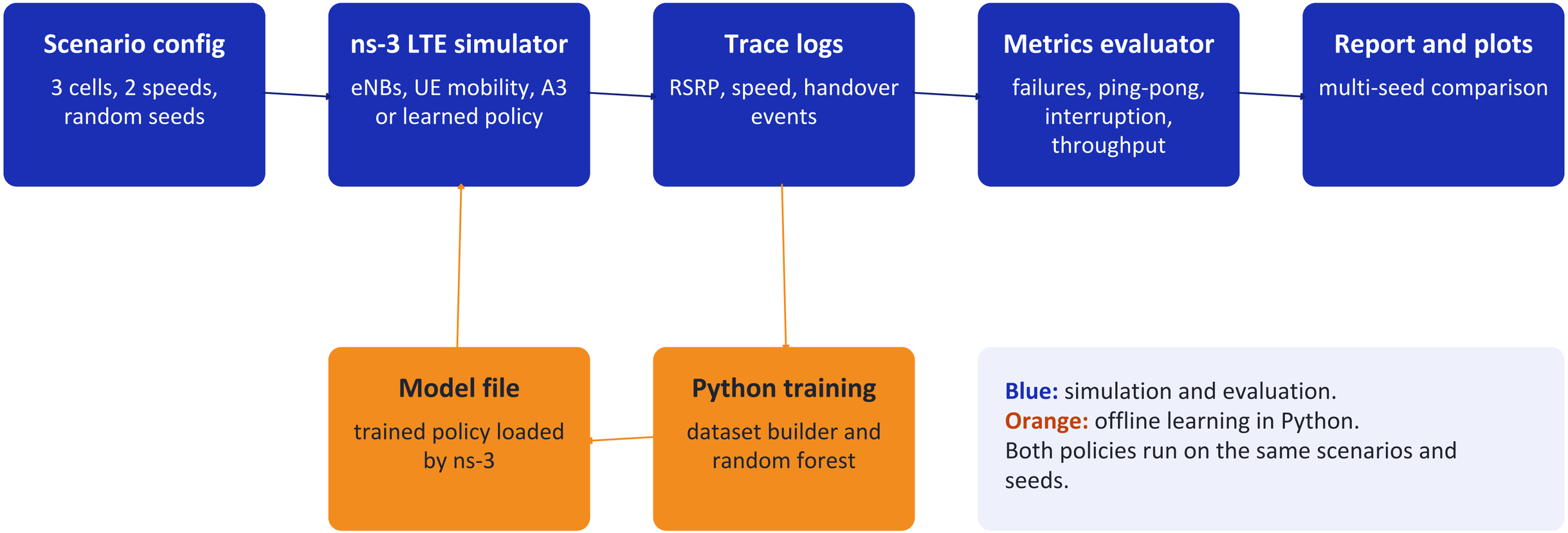
Radio model: ns-3 LTE module with EPC, FDD downlink and RRC-level handover. Planned set-up: 3 eNBs in a line, 10 MHz channel, log-distance path loss. Final parameters will be fixed in the repository config [4].

Optimization Proposed

Aspect	Signal-based baseline (A3)	Learned policy (ours)
Decision input	Serving and neighbour RSRP only	RSRP of cells, RSRP trend, user speed, time since last handover
Speed handling	Same offset and time-to-trigger at every speed	Speed is an input, so decisions can adapt to how fast the user moves
Ping-pong control	Hysteresis and time-to-trigger only	Learns to avoid switching back soon after a handover
Tuning	Parameters set by hand	Learned from simulation data

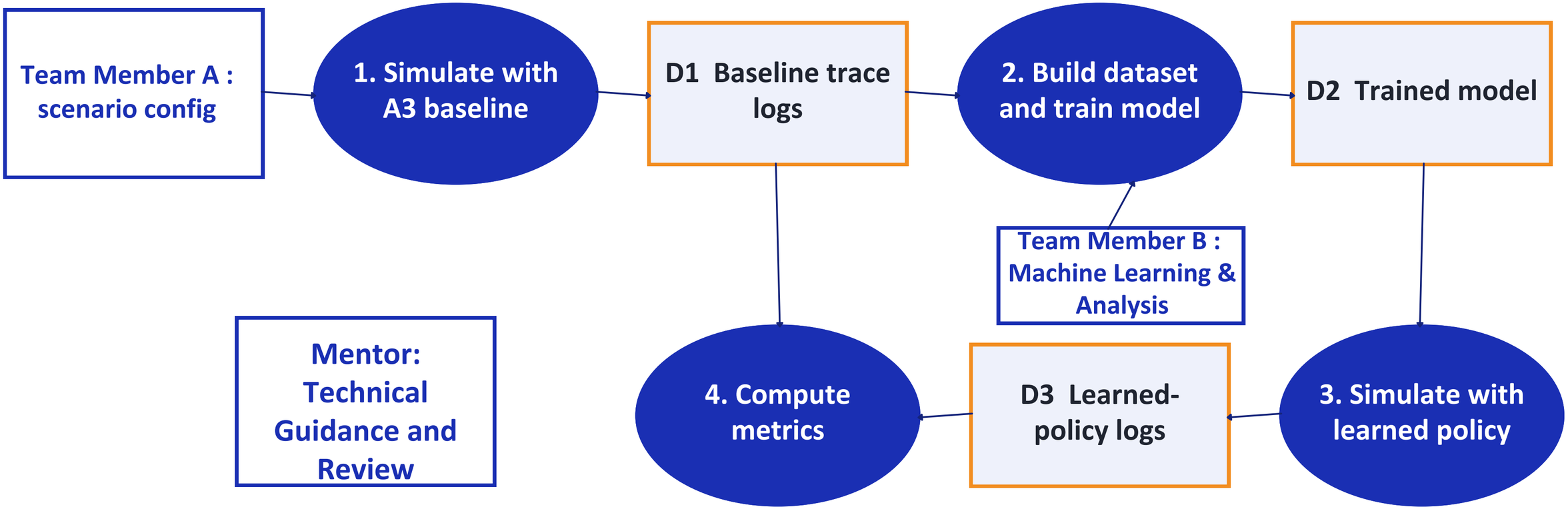
Hypothesis to test - the learned policy lowers handover failures and ping-pong at high speed without losing throughput.

Solution Architecture & Design



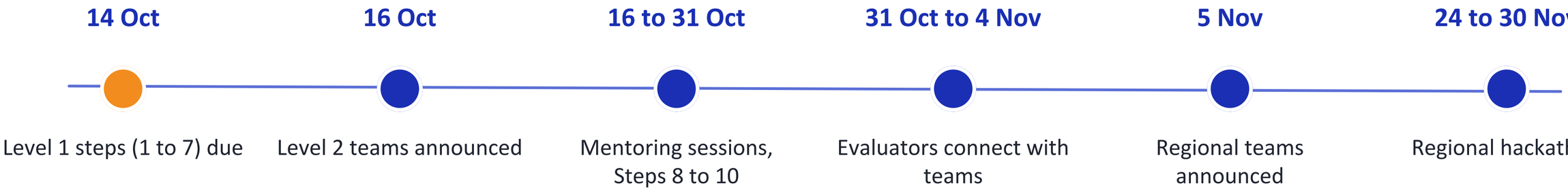
Blue: simulation and evaluation.
Orange: offline learning in Python.
Both policies run on the same scenarios and seeds.

Data Flow Diagram



Deployment Timeline

AIORI-3 milestones



Our project plan

Phase 1 · 16 to 22 Oct

Build

- Install ns-3 and LTE module
- 3-cell scenarios, 30 and 120 km/h
- A3 baseline and trace logging

Phase 2 · 23 to 27 Oct

Learn

- Build dataset from logs
- Train and validate random forest
- Integrate learned policy into ns-3

Phase 3 · 28 to 31 Oct

Evaluate and package

- Multi-seed runs of both policies
- Metrics, plots and final report
- Demo, pseudocode and README

Bibliography

1. V. Paxson, G. Almes, J. Mahdavi and M. Mathis, "Framework for IP Performance Metrics," RFC 2330, IETF, May 1998. https://www.rfc-editor.org/rfc/rfc2330.html

2. 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification," TS 36.331.

3. 3GPP, "E-UTRA and E-UTRAN; Overall description; Stage 2," TS 36.300.

4. ns-3 Network Simulator, "LTE module documentation," nsnam.org.

5. L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

6. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

Thank You!

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